Lecture 3
Describing and Visualizing Distributions

Review

1 What is a statistic?

A numerical characteristic of a sample – a number computed from the observations we actually collected.

2 What is a parameter?

The same kind of number for a population, computed from every observation – and usually unknown.

3 What do we use data for?

To answer questions about a population.

4 What is a population?

The set of all observations of interest – observed and unobserved.

5 What is a sample?

The subset of the population that we actually observe.

Review

1 What do we call the way we collect the data?

The sampling design.

2 What are the three arms of statistics?

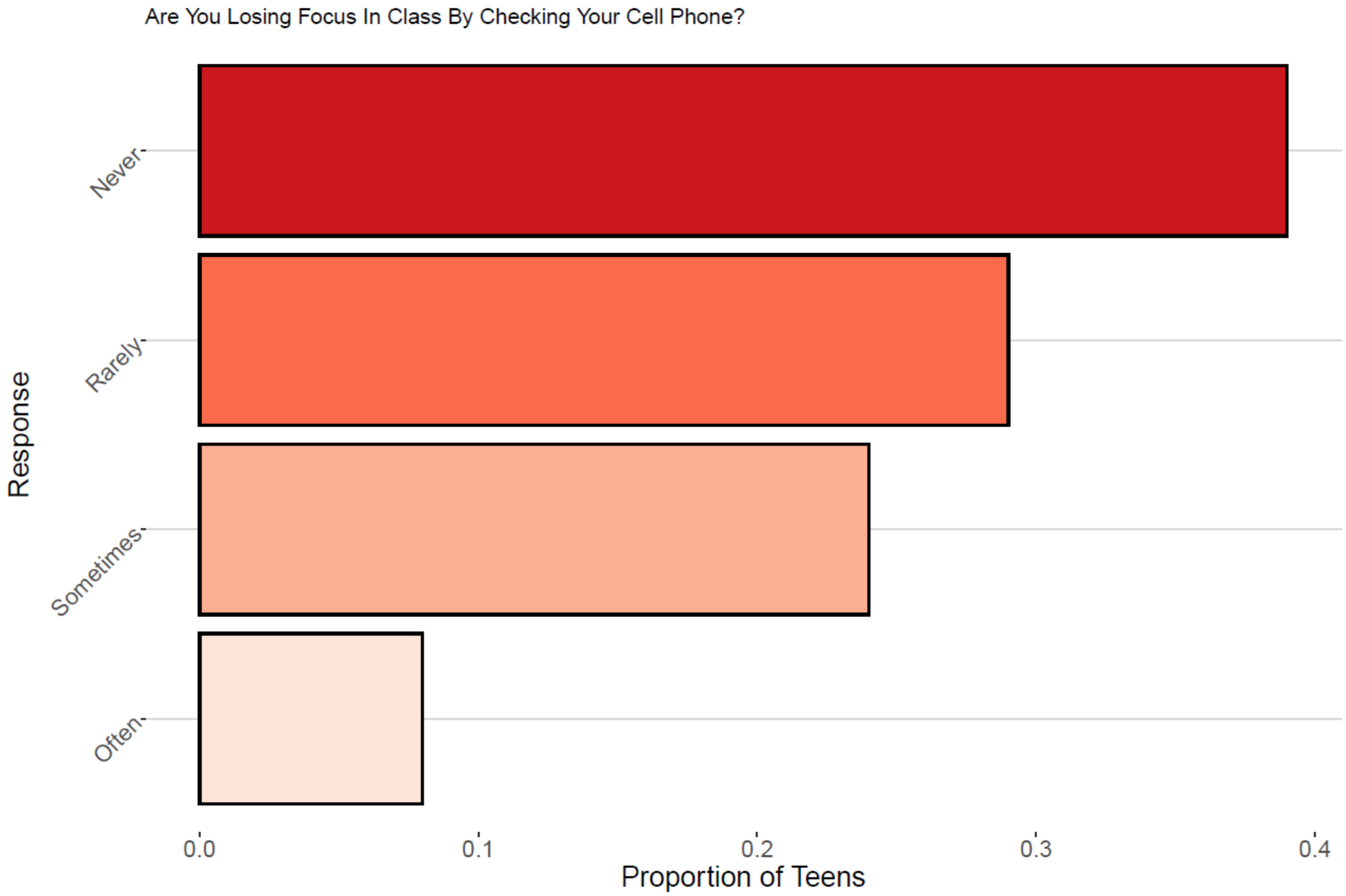
Design, description and inference.

3 Which one comes first, and why?

Design – how we plan to collect the data, settled before any data exists. No amount of description or inference repairs a bad design.

4 What is an estimate?

A statistic computed from a sample, used to approximate an unknown population parameter


Are You Losing Focus In Class By Checking Your Cell Phone?
Response
Never
Rarely
Sometimes
Often
0.0
0.1
0.2
0.3
0.4
Proportion of Teens

Features of Distributions

- A **distribution** of a variable gives (a) the values that occur and (b) how often each value occurs

<u>**Features of A Distribution**</u>

Categorical Variables:

- **Modal category** – the category with the highest frequency

Quantitative Variables:

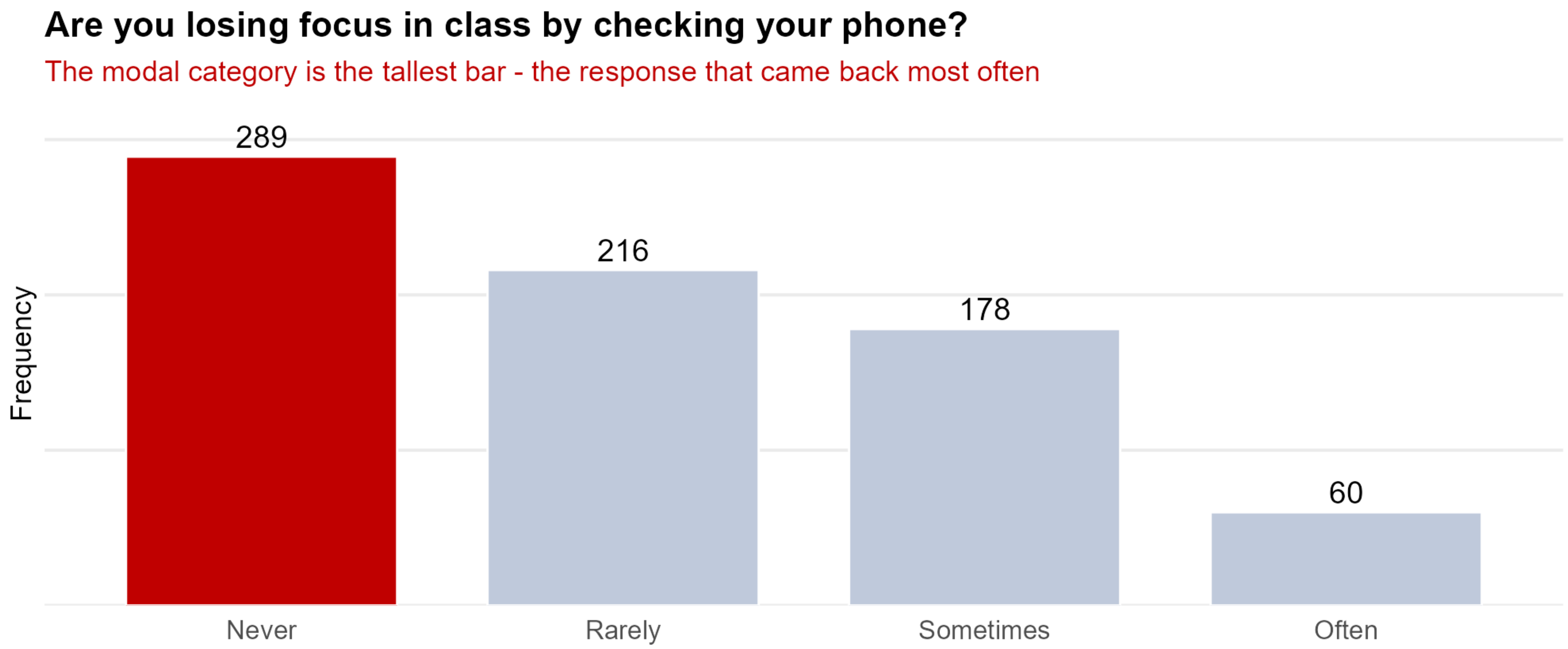
- **Shape** – do observations cluster into certain areas, are values spread out or more densely packed together?

- **Center** – where is the middle point on the distribution, where does a typical value fall?

- **Variability** – how tightly to observations cluster around the center of the distribution

What to Look For: the Modal Category

Modal category – the category with the highest frequency. It is the feature we describe for a qualitative variable.



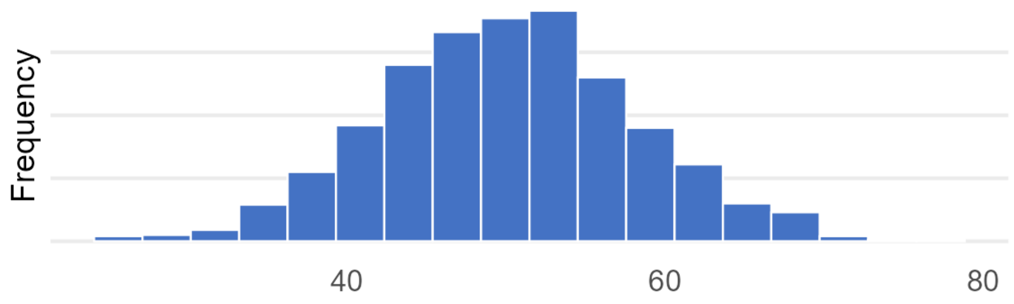
Never is the modal category. There are no numbers to average here, so the modal category is the whole summary – which is why shape, center and variability are reserved for quantitative variables.

What to Look For: Shape

Shape – do observations cluster into certain areas? Are the values spread out evenly, or piled up on one side?

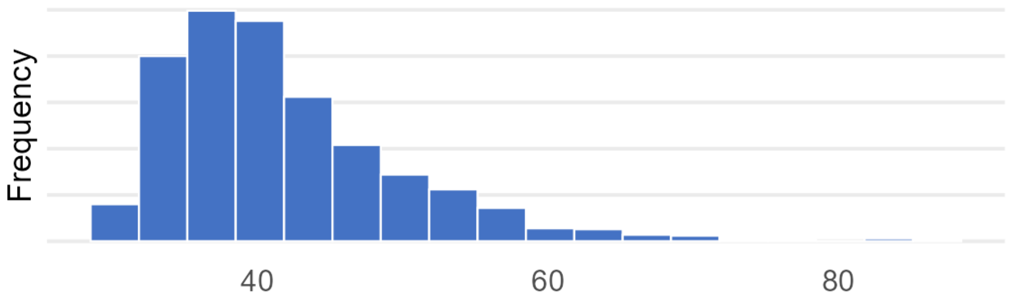
Values pile up in the middle

and thin out evenly on both sides



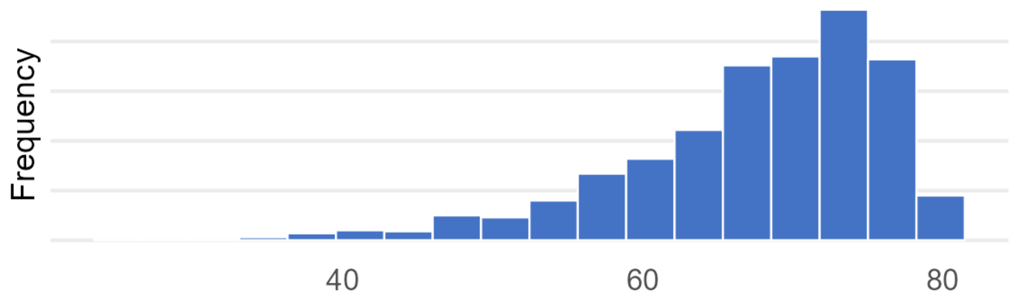
Values pile up on the left

trailing off a long way to the right



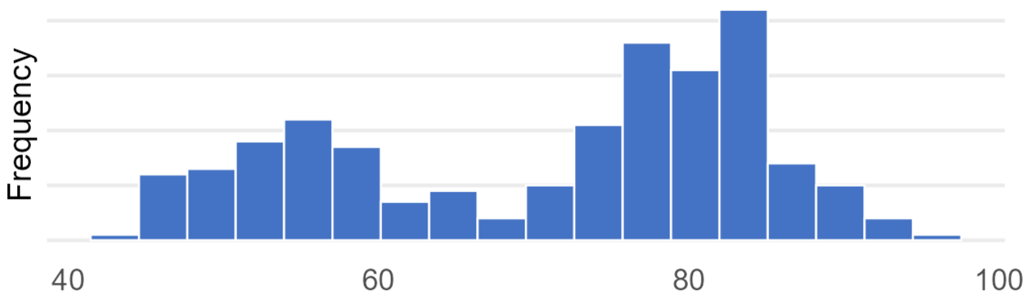
Values pile up on the right

trailing off a long way to the left



Values pile up in two separate places

with a gap between them (Old Faithful)

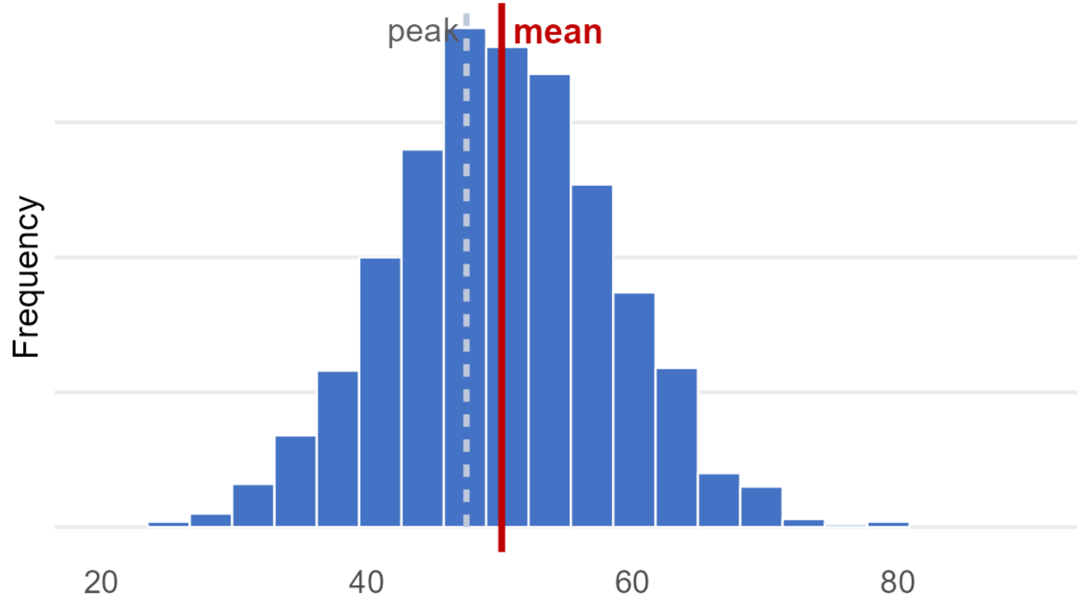


What to Look For: Center

Center – where is the middle of the distribution? Where does a typical value fall?

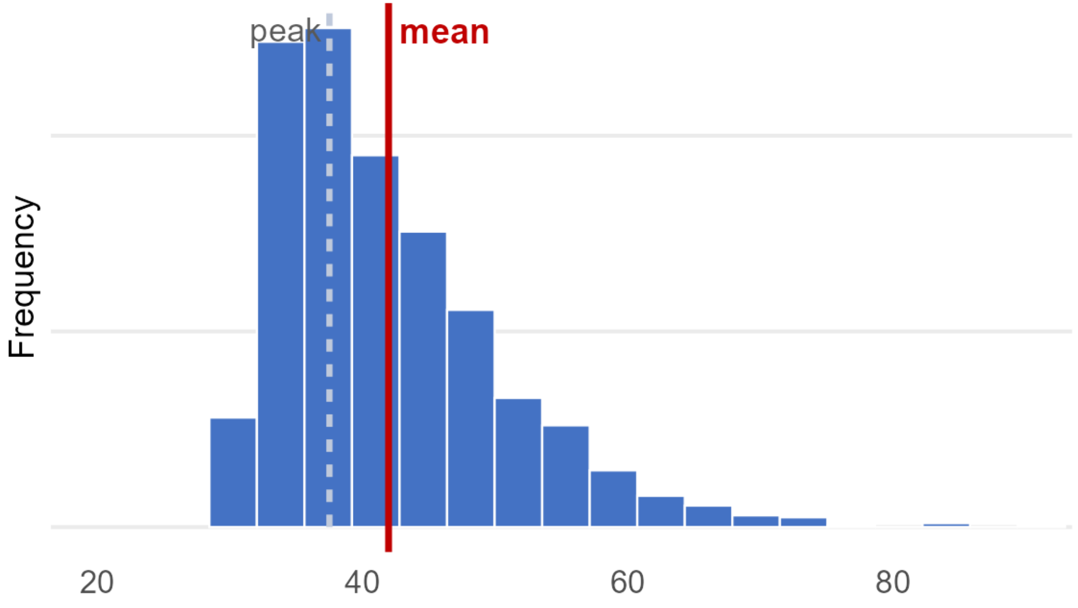
Values pile up evenly

the mean sits under the peak



Values trail off to the right

the long tail drags the mean off the peak



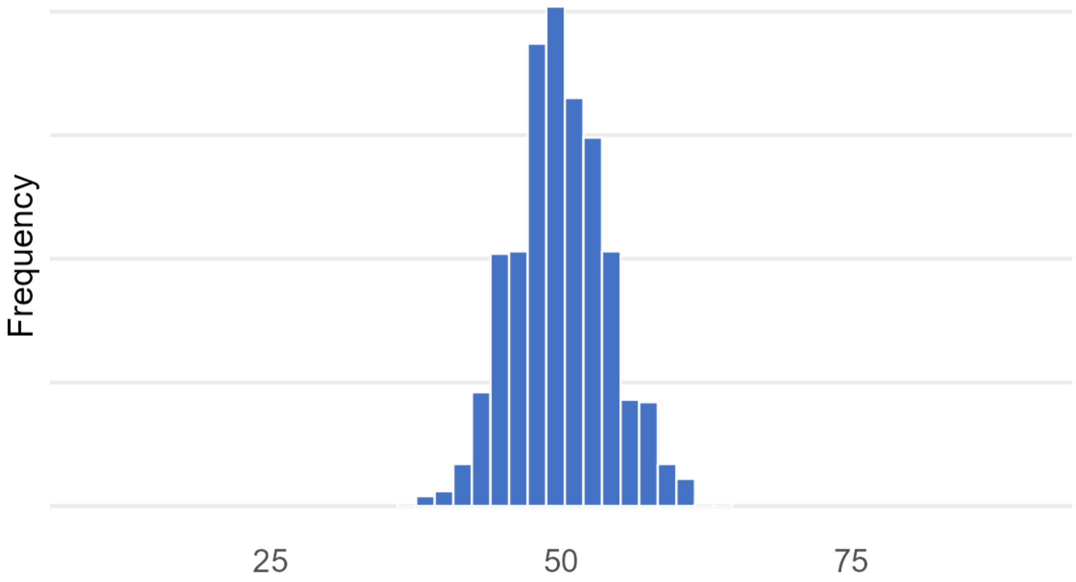
When the values pile up evenly the mean sits under the peak. When they trail off to one side the tail drags the mean away from where most of the data actually are. the mean alone can mislead.

What to Look For: Variability

Variability – how tightly do observations cluster around the center of the distribution?

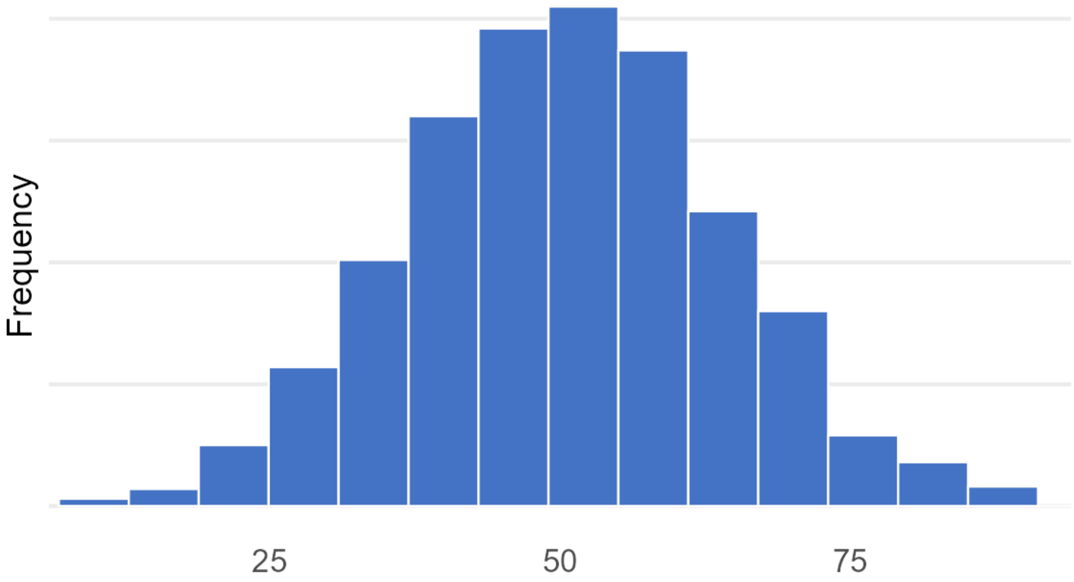
Small variability

values pile up close to the center



Large variability

the same center, spread far wider



Both distributions have the same center, and the axes are identical. Everything that differs between these two pictures is variability: e.g how much the data spread out.

Displaying Distributions: Frequency Tables

- **Frequency table –** a table listing the distinct values of variable together with the number of observations of each value

- **Frequency** – the number of times a value occurs

- **Relative Frequency** – the proportion of observations that assume a given value

$$\text{RF} = \frac{f}{n},$$

- **Cumulative Relative Frequency** - the proportion of observations equal to or less than a given value (more on this later)

Ex. Are teens distracted by their cell phones?

Response	Frequency	Relative Frequency	Cumulative Relative Frequency
Never	289	0.39	0.39
Rarely	216	0.29	0.68
Sometimes	178	0.24	0.92
Often	60	0.08	1.00
Total	743	1.00	

Example: Mendel's Pea Plants

In one of Gregor Mendel's classic studies he bred 8,023 pea plants and recorded the color of the peas – the seeds – that each plant produced.

Pea Color	Frequency	Relative Frequency
Yellow	6022	0.751
Green	2001	0.249

Pea color is set by a single gene.

Yellow (Y) is **dominant**; green (y) is **recessive**. A plant is green only if it inherits y from both parents.

Cross two Yy plants and the offspring are YY, Yy, yY, yy in equal proportion – and only yy shows green. So the theory predicts 3 yellow to every 1 green.

Mendel counted 0.751 and 0.249 – almost exactly 3/4 and 1/4. A frequency table is doing real scientific work here.

Try it out: A Frequency Table by Hand

We put Pew's question to 20 students in this room: are you losing focus in class by checking your phone?

```
Sometimes  Never  Rarely  Never  Often
Rarely  Never  Sometimes  Rarely  Never
Sometimes  Rarely  Never  Often  Sometimes
Rarely  Never  Rarely  Sometimes  Never
```

Build the frequency table: frequency, relative frequency and cumulative relative frequency. Which is the modal category?

Response	f	Rel. freq.	Cum. rel. freq.
Never	7	0.35	0.35
Rarely	6	0.30	0.65
Sometimes	5	0.25	0.90
Often	2	0.10	1.00
Total	20	1.00	

Modal category: Never. Pew's 743 teens gave 0.39, 0.29, 0.24, 0.08 – 20 students landed within a few points of them.

Try it out: A Discrete Variable

Ten students report how many siblings they have. Writing the responses as a set:

```
R = {0, 1, 1, 2, 0,
        3, 1, 2, 1, 4}
```

Build the frequency table. Which value comes up most often? Where do the values pile up, and how do they trail off?

R	f	Rel. freq.
0	2	0.20
1	4	0.40
2	2	0.20
3	1	0.10
4	1	0.10
Total	10	1.00

The most common value is 1, but the mean is 1.5: the few large values pull the mean above where the bulk of the data sit – the same effect you saw on the center slide.

Frequency Tables for Continuous Variables

- The number of possible values is usually very large
- Convert continuous values into discrete groups (sometimes called bins):

Steps:

1. Divide the range of the variable into a set of non-overlapping intervals
2. Count the number of values that fall into each interval

Try it out: Binning a Continuous Variable

Fifteen observations of a continuous variable, written as a set:

```
X = {1.23, 1.38, 1.76, 2.60, 2.65,
        2.79, 2.89, 3.12, 3.15, 3.18,
        3.65, 3.77, 3.87, 4.12, 4.41}
```

No two values are alike, so listing them gives 15 values of frequency 1 and tells us nothing. Group them into whole-number bins.

Bin width is a choice, not a property of the data. Halving it splits the middle and exposes an empty bin: the values fall into two clusters.

Next: the same procedure on 272 Old Faithful eruptions.

Bin	f	Rel. freq.
$1 \le X < 2$	3	0.20
$2 \le X < 3$	4	0.27
$3 \le X < 4$	6	0.40
$4 \le X < 5$	2	0.13
Total	15	1.00

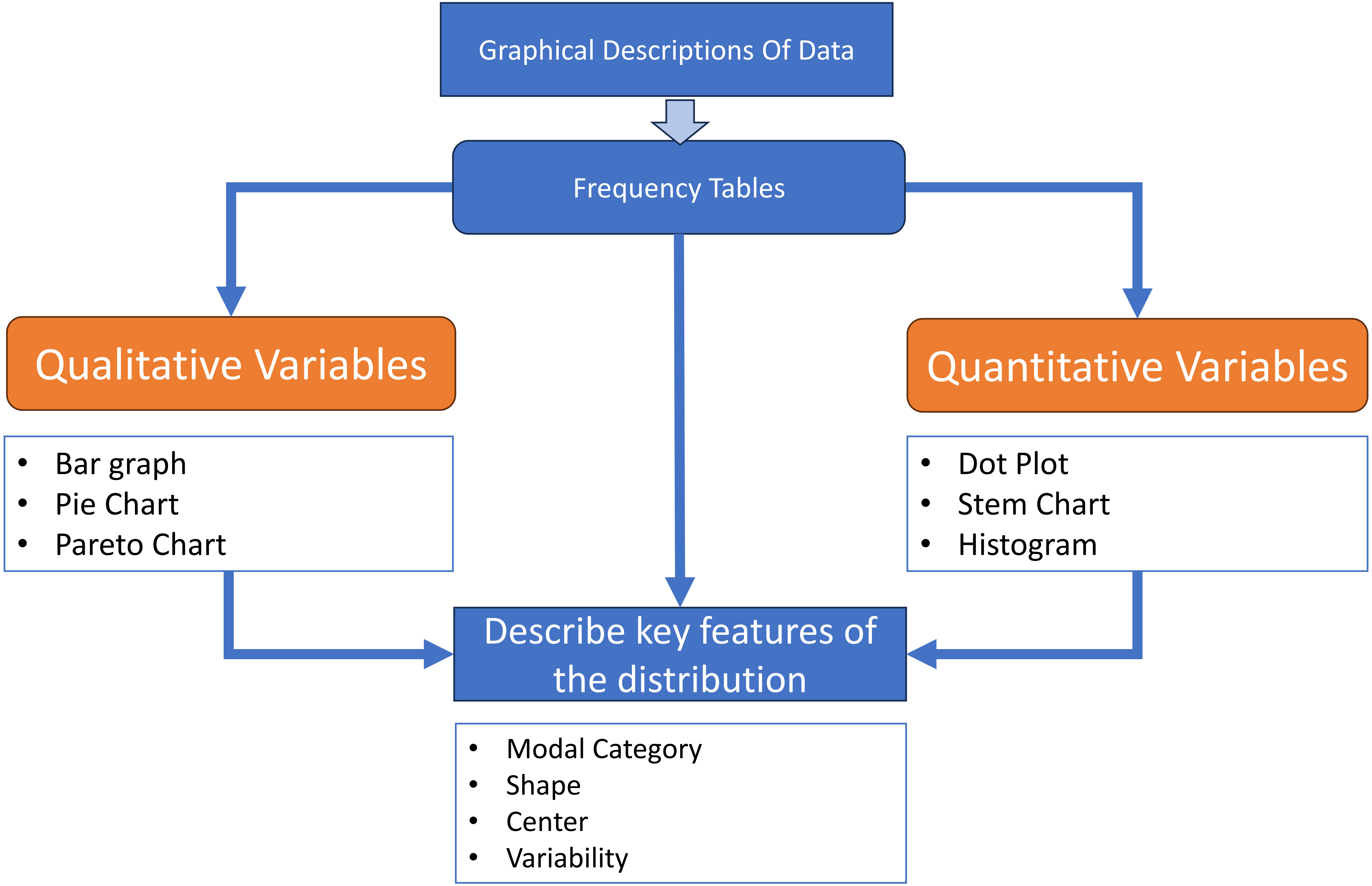
Narrower bins	f	Rel. freq.
$1.0 \le X < 1.5$	2	0.13
$1.5 \le X < 2.0$	1	0.07
$2.0 \le X < 2.5$	0	0.00
$2.5 \le X < 3.0$	4	0.27
$3.0 \le X < 3.5$	3	0.20
$3.5 \le X < 4.0$	3	0.20
$4.0 \le X < 4.5$	2	0.13

Example: Old Faithful Eruption Times

Observation	Eruption Time	Waiting Time
1	3.600	79
2	1.800	54
3	3.333	74
4	2.283	62
5	4.533	85
6	2.883	55
7	4.700	88
8	3.600	85
9	1.950	51
10	4.350	85
11	1.833	54
⋮	⋮	
272	4.467	74

Old Faithful Eruption Times: Frequency Table

Eruption Duration (Min)	Frequency	Relative Frequency	Cumulative Relative Frequency
$1.5 \leq X < 2.0$	55	0.20	0.20
$2.0 \leq X < 2.5$	37	0.14	0.34
$2.5 \leq X < 3.0$	5	0.02	0.36
$3.0 \leq X < 3.5$	9	0.03	0.39
$3.5 \leq X < 4.0$	34	0.12	0.51
$4.0 \leq X < 4.5$	75	0.28	0.79
$4.5 \leq X < 5.0$	54	0.20	0.99
$5.0 \leq X < 5.5$	3	0.01	1.00

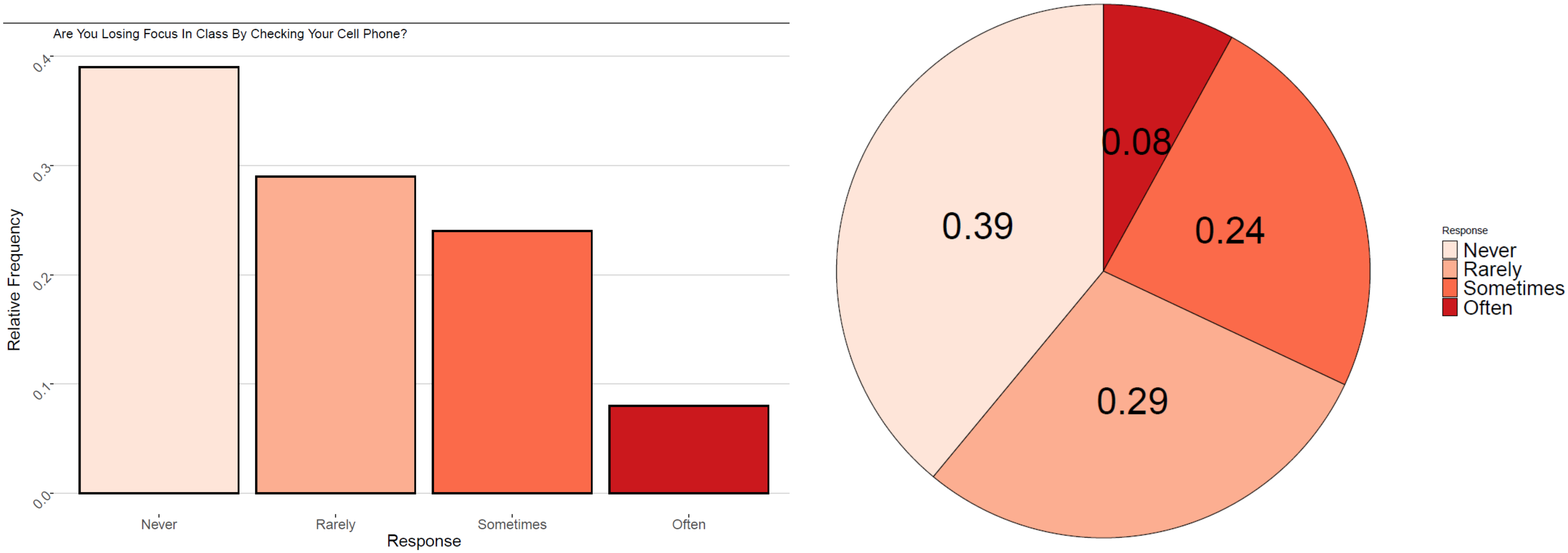

Graphical Descriptions Of Data
Frequency Tables
Qualitative Variables
Quantitative Variables
• Bar graph
• Pie Chart
• Pareto Chart
• Dot Plot
• Stem Chart
• Histogram
Describe key features of the distribution
• Modal Category
• Shape
• Center
• Variability

Visualizing Distributions of Categorical Data

Pie Charts - a circle divided into 'slices' corresponding to each category. The size of a slice shows the proportion of observations in a category

Bar graph – displays a vertical bar for each category. The height of the bar shows the percentages of observations in the category

Pareto Chart - a bar chart with the categories ordered by decreasing frequency



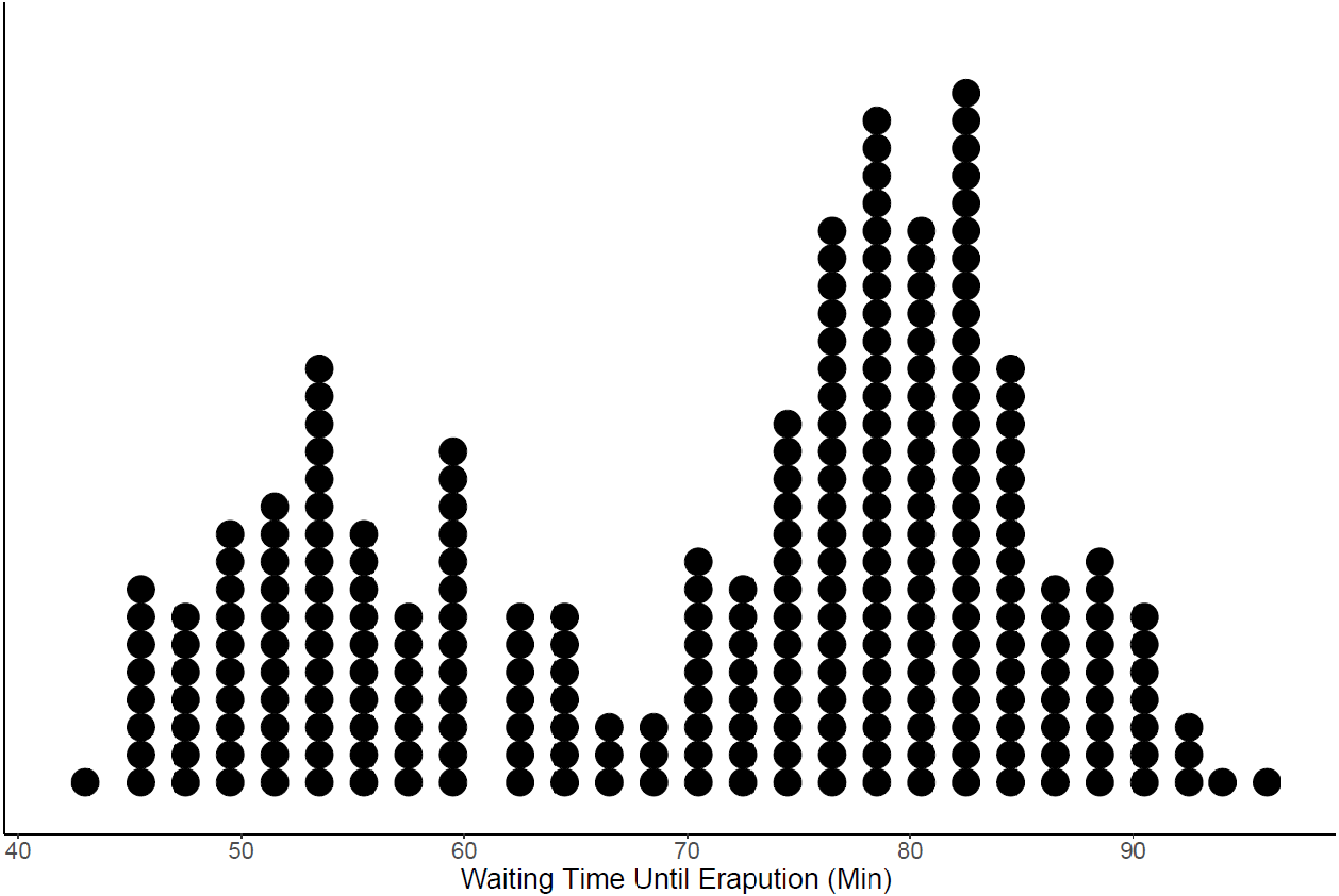
Are You Losing Focus In Class By Checking Your Cell Phone?
Relative Frequency
0.4
0.3
0.2
0.1
0.0
Never
Rarely
Sometimes
Often
Response
0.08
0.39
0.24
0.29
Response
Never
Rarely
Sometimes
Often

Visualizing Distributions: Quantitative Variables

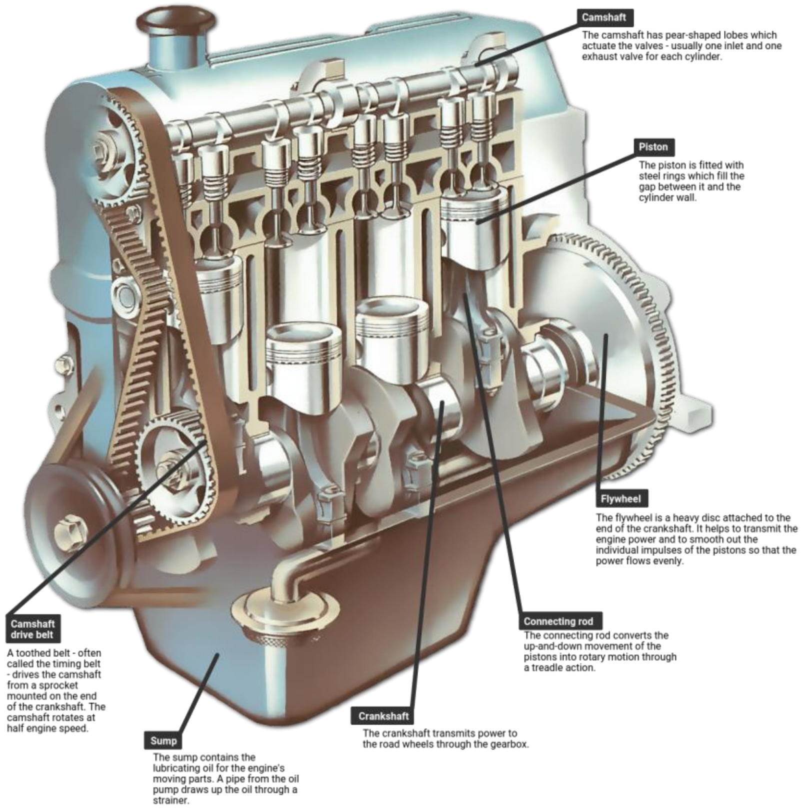
- **Dot plots –** shows a dot for each observation placed above the value for that observation

Steps to construct a dot plot

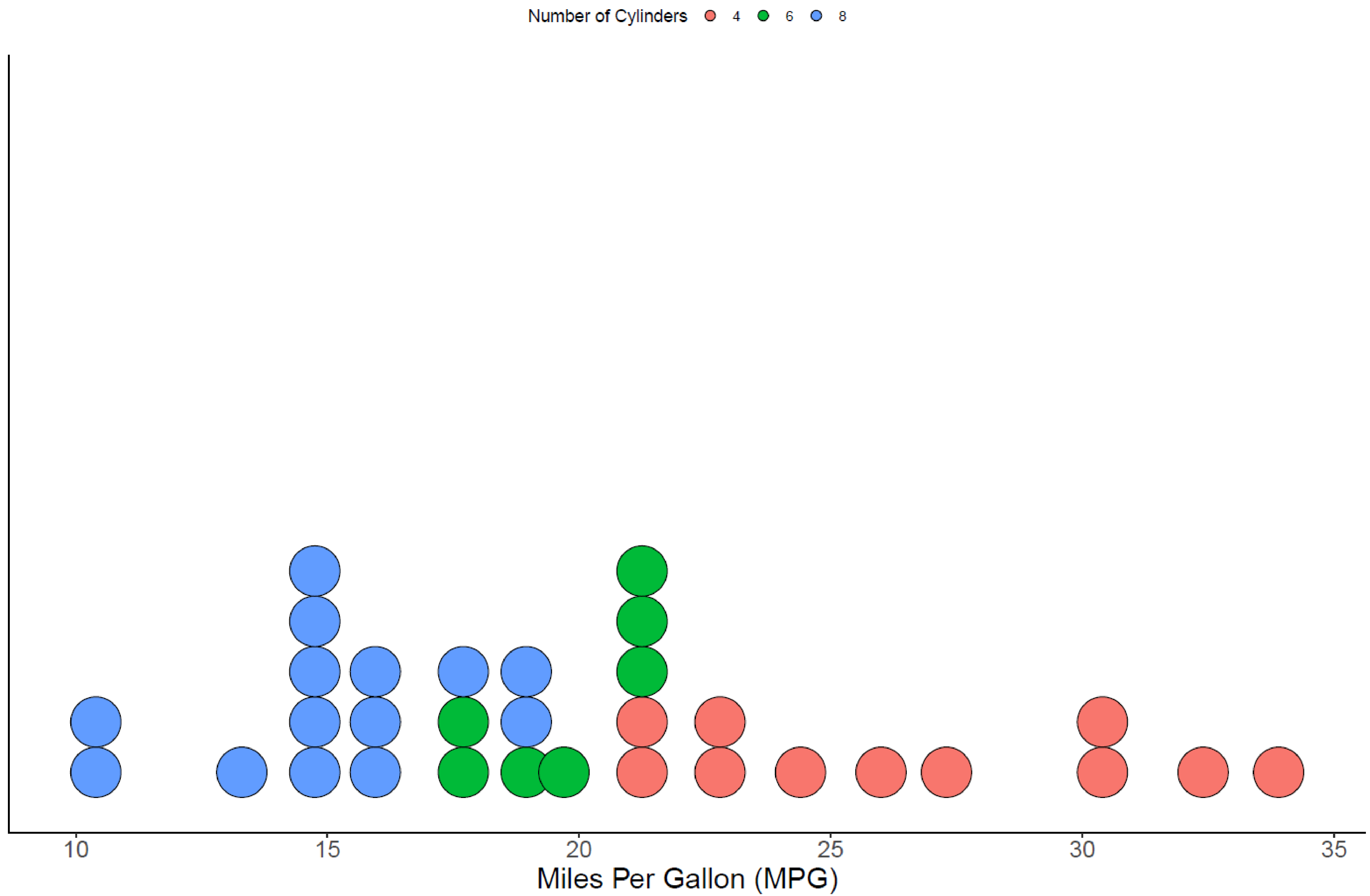
1. **Draw a horizontal line and mark the line with regular values of the variable**
2. **For each observation, place a dot above its value on the number line**
   - Works best with quantitative discrete data
   - Doesn’t work well if the variable is continuous and takes on many distinct values…
   - For continuous data, the values may need to be round to the nearest tenth or integer


40
50
60
70
80
90
Waiting Time Until Eraption (Min)

Example: MPG and Engine Cylinders



Observation	MPG	Cylinders	Model
1	21.0	6	Mazda RX4
2	21.0	6	Mazda RX4 Wag
3	22.8	4	Datsun 710
4	21.4	6	Hornet 4 Drive
5	18.7	8	Hornet Sportabout
6	18.1	6	Valiant
$\vdots$	$\vdots$	$\vdots$	$\vdots$
32	21.4	4	Volvo 142E


Number of Cylinders
4
6
8
10
15
20
25
30
35
Miles Per Gallon (MPG)

Visualizing Distributions: Quantitative Variables

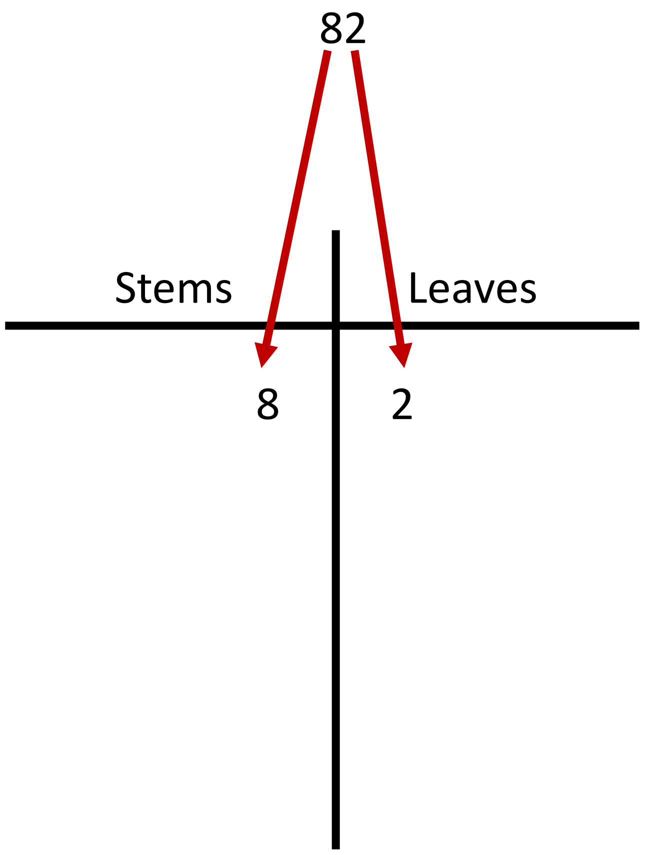
Stem and leaf plot – like a dot plot, a stem and leaf diagram also displays individual observations.

Stem – all the digits in an observation except the last digit

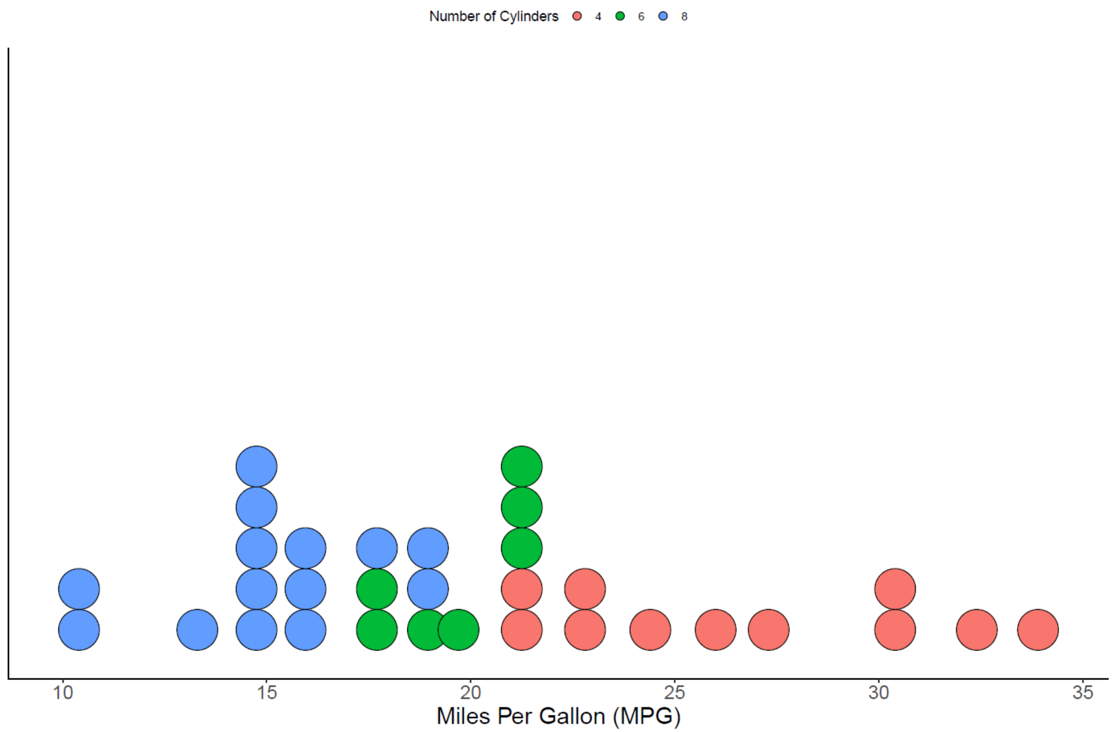
Leaf – the last digit in an observation

Steps to construct a stem and leaf plot

1. **Sort the data in order from smallest to largest.**
2. **Place the stems in a column in increasing order**
3. **Place a vertical line to the right of the stems**
4. **To the right of the vertical line, fill in the leaves that correspond with each stem in increasing order**



Example: MPG



Observation	MPG	Cylinders	Model
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4	21.4	6	Hornet 4 Drive
5	18.7	8	Hornet Sportabout
6	18.1	6	Valiant
⋮	⋮	⋮	⋮
32	21.4	4	Volvo 142E

Stems	Leaves
10	4,4
11	
12	
13	3
14	3,7
15	0,2,2,5,8
16	4
17	3,8
18	1,7
19	2,2,7
20	
21	0,0,4,4,5
22	8,8
23	
24	4
25	
26	0
27	3
28	
29	
30	4,4
31	
32	4
33	9

Stems	Leaves
10	4,4
12	3
14	3,7,0,2,2,5,8
16	4,3,8
18	1,7,2,2,7
20	0,0,4,4,5
22	8,8
24	4
26	0,3
28	
30	4,4
32	4,9

Try it out: Stem and leaf plot

Data = 4.2, 3.8, 4.6, 3.2, 2.7, 8.2, 9.1, 0.2, 1.2, 6.2

Visualizing Distributions: Quantitative Variables

Stem and leaf plots and **dot plots** are unwieldy for large n

Histogram – uses bars to portray the frequencies or relative frequencies of the possible outcomes for a quantitative variable

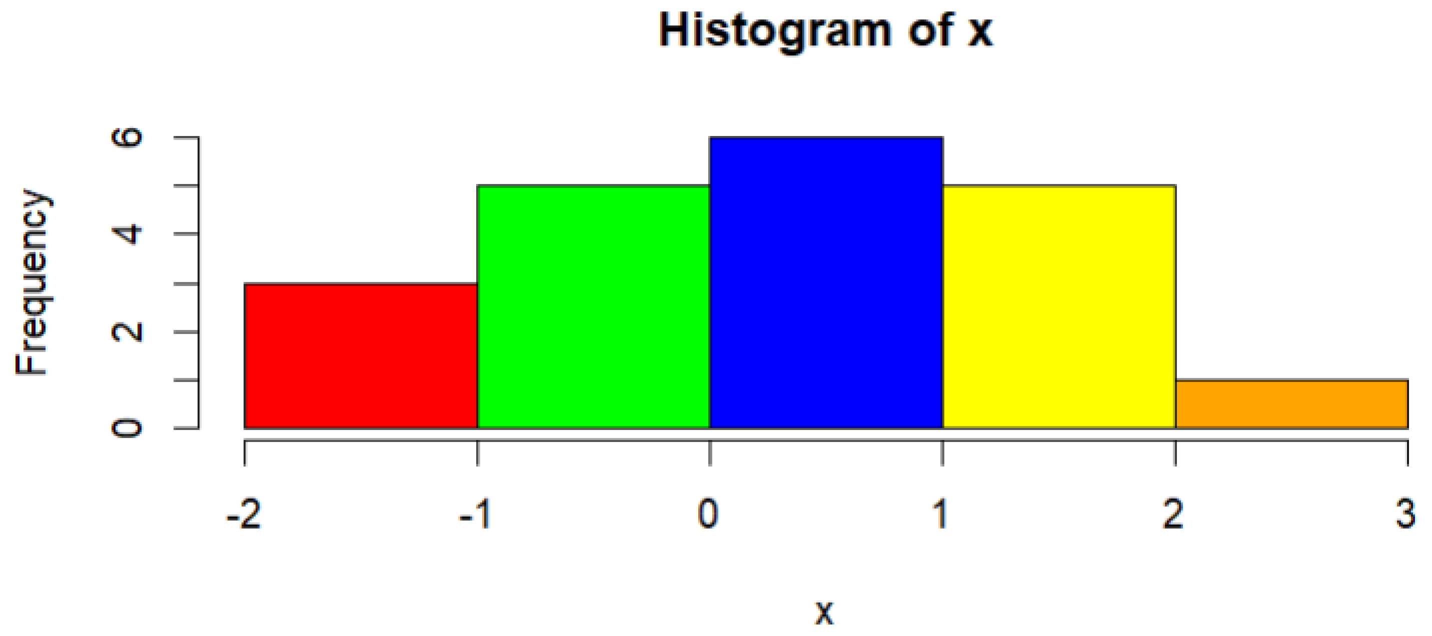
Steps to construct a histogram

1. **Divide the range of the data into intervals of equal width**
2. **Compute the frequency of each interval (i.e construct the frequency table)**
3. **Label the x-axis with the values or endpoints of each interval.**
4. **Draw a bar over each value or interval with height equal to its frequency or relative frequency**

Try it out: Histogram

Consider the following $n = 20$ observations of a continuous variable

	Bin 1	Bin 2	Bin 3	Bin 4	Bin 5
x =	-1.5, -1.2, -1.0,	-0.8, -0.7, -0.6, -0.1, -0.1,	0.1, 0.1, 0.1, 0.6, 0.6, 0.8,	1.1, 1.2, 1.3, 1.8, 1.9,	2.4



How to choose the number of Bins?

- How to choose the best number of bins is not a straightforward question and there is a lot of literature on the subject
- We can construct our histogram using a specific binwidth w or under a set number of bins k
- $w = \dfrac{\max x - \min x}{k}$ or $k = \dfrac{\max x - \min x}{w}$
- Square root method: $k = \mathrm{round}(\sqrt{n})$ (A fairly safe and basic rule of thumb)
- Sturges Rule[1]: $k = \mathrm{round}(\log_2 n) + 1$ (not great for $n < 30$)
- Rices Rule[2]: $k = 2\sqrt[3]{n}$

[1] Sturges, Herbert A. "The choice of a class interval." Journal of the american statistical association 21.153 (1926): 65-66.

[2] Lane, David. *Online statistics education: A multimedia course of study*. Association for the Advancement of Computing in Education (AACE), 2003. – Chapter 2 “Graphing Distributions

Some tips

- If <u>too few</u> intervals are used, then the graph will be too crude

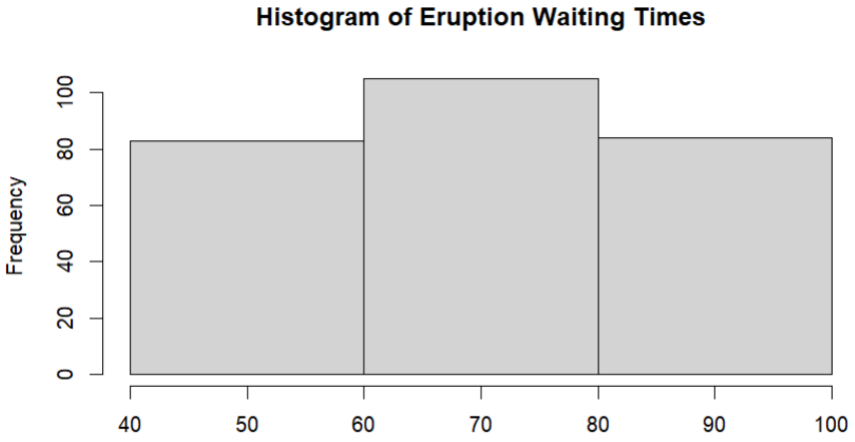
- If <u>too many</u> intervals are used, graph will contain many short bars and gaps.

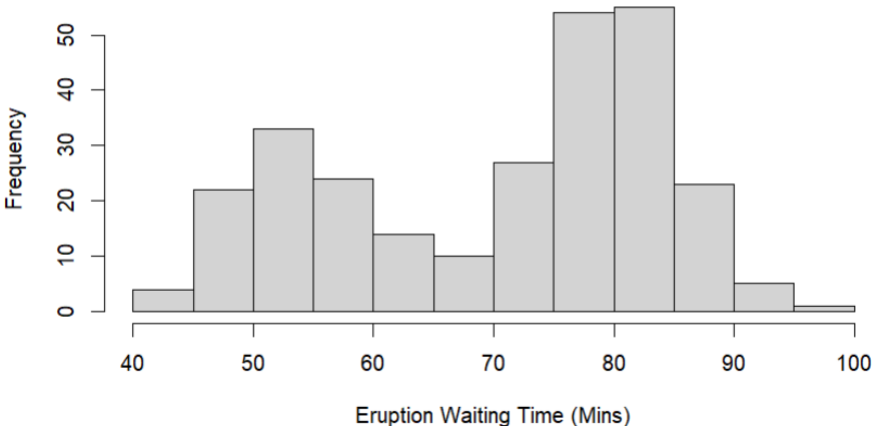
  Usually between 5 - 15 intervals are enough.

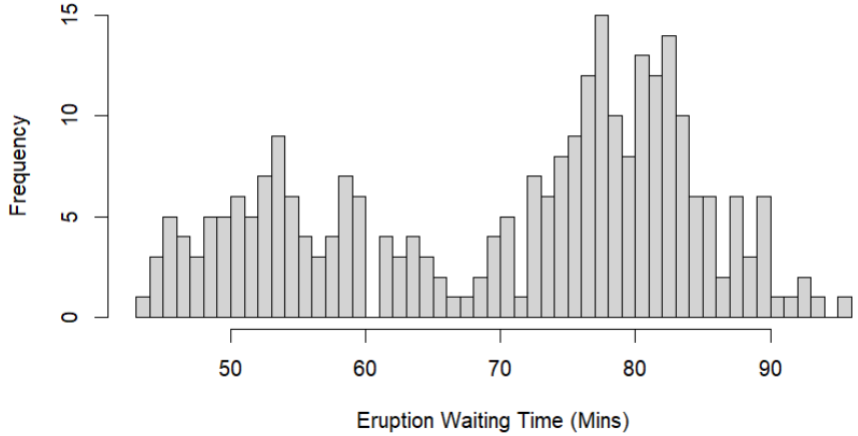
- Most plotting software will automatically choose the number of bins.

- **<u>ALWAYS</u>** plot the histogram to get an idea about the shape of the distribution of a quantitative variable

- Is the number of observations is small (say n < 50) then it's a good idea to supplement a histogram with a dot plot or stem plot







Example: Old Faithful Eruption Times

Waiting Time (Min)	Frequency	Relative Frequency	Cumulative Relative Frequency
< 50	21	0.077	0.077
50 - 60	56	0.206	0.283
60 - 70	26	0.096	0.379
70 - 80	77	0.283	0.662
80 - 90	80	0.294	0.956
> 90	12	0.044	1